

# Engineering drawings a model can draw but cannot always stand behind

Motivation, problem statement, architecture, observed failures

## 1 Motivation

A structural drawing is a contract. The dimensions assert lengths, the schedule asserts sections, and the results panel asserts what the structure does under load. Downstream, somebody sizes a member or signs off a design because of the number in that panel.

A language model can now draft these drawings to a standard that looks professional: correct projection, dimension lines with extension witnesses, a member schedule, load arrows, support symbols, a modelling-assumptions note. Figure 2 is one, produced in a single call. That quality is exactly what makes it dangerous, because the appearance of the drawing carries no information about whether the number in its results panel is right. Nothing on the page distinguishes a solved value from a plausible one.

The only way to tell is to re-solve the structure independently and compare. This brief reports what that comparison found.

## 2 Problem statement

**When a model generates or edits an engineering drawing, can the numerical claims printed on it be trusted — and if not, where and how do they break?**

The object of study is a dimensioned structural elevation of a physical assembly — a building frame, a worktable, a shelf — emitted as a single SVG, together with a JSON analysis block. Editing matters as much as generating: moving a column grid or changing a load invalidates every dimension and every derived quantity at once, while leaving the drawing that carries them looking untouched.

Three outcomes must be held apart, because they have different consequences:

### Correct

the claims agree with an independent solve.

### Wrong

a number is printed and it is false. A released drawing now carries a false engineering claim, and nothing on the page reveals it.

### Declined

the model states it did not perform the analysis and returns no numbers. The deliverable is incomplete, but honest.

## 3 Architecture

The job is stated as two arms, and both are checked the same way.

**text  $\rightarrow$  SVG**

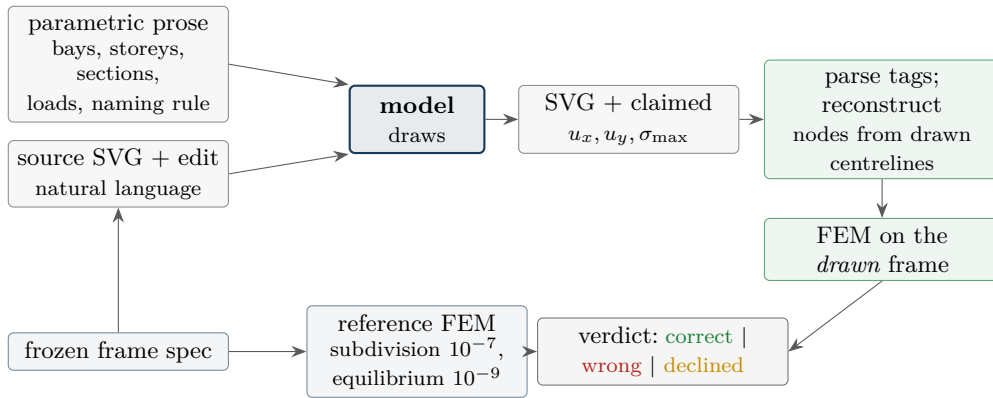
a parametric prose description of a frame — bays, storeys, spacings, setback rule, sections by role, loads by position, and a naming convention — and the model must derive every node coordinate and every member before drawing. No structured model is supplied.

**SVG  $\rightarrow$  SVG**

an existing drawing plus a natural-language edit. The model must recover the geometry from the drawing itself, apply the edit, and redraw. No structured model here either.

Verification does not rest on the returned numbers. The scorer parses the tagged geometry, reconstructs node positions from the *drawn* member centrelines, and re-solves that frame with the same linear frame FEM that built the reference. Two comparisons follow: the claimed analysis against the reference, and the physics of the drawing itself against the reference.

The second needs stating precisely, because the shared scorer computes that re-solve and records it *without* letting it decide the outcome — its verdict rests on member coordinates, dimensions and the claimed analysis. A separate regrade pass makes the re-solve decide as well, so a drawing whose own physics disagrees with the reference cannot pass on coordinates alone. That pass can only withdraw a pass, never grant one. Applied to the 33 samples already measured it changes nothing: the worst disagreement between the drawn frame and its reference is  $3 \times 10^{-10}$ , seven orders inside the  $2 \times 10^{-3}$  gate. The same gate is enforced when generating training targets.



**Figure 1:** Both arms feed one verifier. The frozen specification builds the reference and the source drawing, but never reaches the model; agreement is therefore evidence rather than construction. The deterministic exporter that draws from the specification is the oracle — its own drawings pass every check by construction, which is the harness’s positive control.

Two consequences follow. The exporter path means a correct drawing can always be produced from a solved specification without a model at all, so the model is only ever being asked whether it can reach the same place unaided. And because the checks run on reconstructed geometry, a drawing that merely copies its input fails the edited-frame FEM outright — which is the defence against the copy-reward trap that makes similarity metrics useless on edit tasks.

**Where Astra sits.** Astra appears in this brief only as the *subject under test*. It is run end-to-end against both arms with no tools, to establish what a frontier model reaches unaided and therefore what the pipeline has to supply for itself. It is not a component of Figure 1, and nothing here is designed around continued access to it.

## 4 Dataset

Everything is generated; no real-drawing corpus is involved. Frames are ordered by degree of static indeterminacy,  $i = 3m + r - 3j$  for  $m$  members,  $r$  reaction components and  $j$  joints, because

a determinate frame yields to statics alone while an indeterminate one needs the simultaneous stiffness system — the part that must be carried out unaided.

| Set                    | Path                    | Tasks | Input                       | Role             |
|------------------------|-------------------------|-------|-----------------------------|------------------|
| text $\rightarrow$ SVG | data/eng-svg-bench-text | 24    | prose, 2.2–2.7 kB           | <b>benchmark</b> |
| SVG $\rightarrow$ SVG  | data/eng-svg-bench-edit | 24    | drawing + edit, 3.7–11.7 kB | <b>benchmark</b> |
| pilot                  | data/cad-astra-pilot    | 6     | JSON model                  | source of case A |
| frames                 | data/eng-frame-bench    | 18    | JSON model                  | source of case B |
| frames XL              | data/eng-frame-bench-xl | 6     | JSON model                  | source of case B |

The two benchmark arms span 3 to 21 members and  $i = 3$  to 27. **Neither supplies a structured model**, which is what separates them from the three lower sets: those handed the model over as JSON, and the measured results in Sections 5 and 6 were all obtained under that easier condition. A record earned with coordinates supplied does not transfer to a task where they must be derived, so the two arms need measuring from scratch.

Each task carries the frozen physical model, the prompt, ground-truth member lengths, the FEM reference (displacements, peak stress, per-member stresses, reactions, equilibrium residual), the deflection and stress limits, the physical-to-SVG mapping, a model hash, and the exporter’s own reference drawing. A reference is admitted only after member-subdivision invariance at  $10^{-7}$  and a dimensionally normalised equilibrium residual below  $10^{-9}$  — the force and moment parts of that residual carry different units and need separate scales, or a large frame is rejected for being large rather than wrong. Nine checks pass on each arm, including that the exporter drawing scores as a pass and that FEM of its reconstructed geometry matches the reference.

Each description ends with a checksum — “*derived correctly this frame has exactly 8 nodes and 8 members*” — so a mis-derivation is detectable rather than silent. The prose is templated, so a systematic failure warrants inspecting the description for ambiguity before it is attributed to the model.

## 5 Failure case A: a correct drawing carrying a false claim

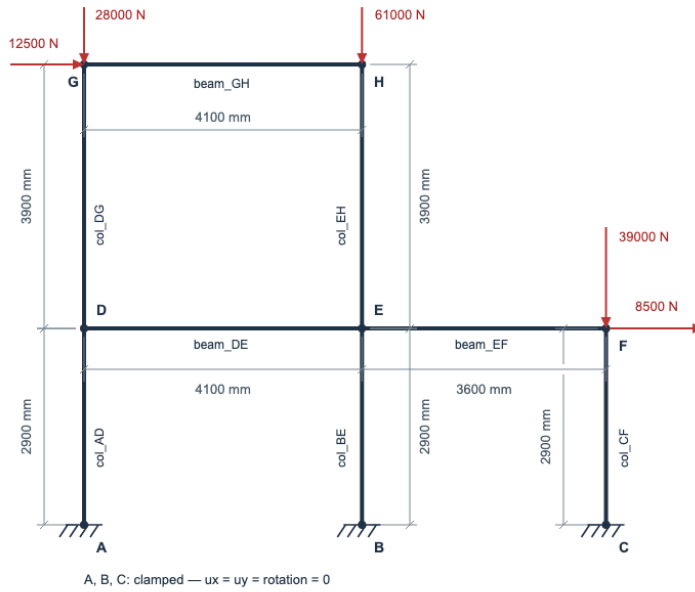
The first case is the one the verification layer exists for: everything a reader can see is right, and the conclusion is wrong.

The task was an *edit*. The model received the original two-storey setback frame, its SVG, and an instruction to move the middle column grid from  $x = 3400$  to  $x = 4100$  mm, raise the upper floor from  $y = 6200$  to  $y = 6800$  mm, and change the vertical load at node H to  $-61$  kN — then to recompute every dimension and every derived quantity. It did all of that in a single call with no tools. The geometry check passes at 1 mm, every joint closes, all eight member lengths are correct, and the visible SVG results agree with the returned JSON. By every structural and drafting criterion the drawing is sound.

### Two-storey setback building frame elevation

Edited geometry • centreline dimensions in mm • undeformed elevation

Middle grid  $x = 4100$ ; upper floor  $y = 6800$ ; H vertical load =  $-61000$  N



#### MEMBER SCHEDULE

Solid rectangles:  $b \times h$ , mm  
h in frame plane; b out of plane

|         |           |
|---------|-----------|
| col_AD  | 160 × 200 |
| col_BE  | 160 × 200 |
| col_CF  | 140 × 180 |
| beam_DE | 140 × 260 |
| beam_EF | 140 × 260 |
| col_DG  | 140 × 180 |
| col_EH  | 140 × 180 |
| beam_GH | 120 × 240 |

All members:  $E = 200000$  N/mm<sup>2</sup>

#### ESTIMATED FRAME RESPONSE

Probe H • +x right, +y up

ux at H: +4.47 mm  
uy at H: -0.0830 mm  
Peak end stress: 179 MPa  
 $|uy| \leq 10.0$  mm: PASS (estimated)  
**Peak stress  $\leq 160$  MPa: FAIL (estimated)**  
Governing end: col\_EH at E

MODEL: 2D linear elasticity; small displacements; rigid joints; axial + Euler-Bernoulli bending.

Nodal loads only. No shear deformation, self-weight, buckling, out-of-plane effects or connection flexibility.

Stress measure: max over member ends of  $|N|/A + |M|(h/2)/I$ .

Results are hand stiffness estimates with axial-compatibility corrections, not a numerical solver result.

Named checks apply only to this idealised frame; no claim of complete building or product safety.

**Figure 2:** building\_edit, 20 September pilot, unmodified output. The edit — move the middle grid to  $x = 4100$ , raise the upper floor to  $y = 6800$ , change the load at H to  $-61$  kN — was applied correctly. Geometry and all eight member dimensions pass at 1 mm. The defect is confined to the results panel.

#### ESTIMATED FRAME RESPONSE

Probe H • +x right, +y up

ux at H: +4.47 mm  
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**Peak stress  $\leq 160$  MPa: FAIL (estimated)**  
Governing end: col\_EH at E

**Figure 3:** Detail of the same panel. Both displacements are right to 0.3%. The peak stress is **179 MPa** against a reference of **20.356 MPa**, a factor of 8.8 — and the drawing therefore declares the frame to *fail* its 160 MPa limit when its true peak is one eighth of it.

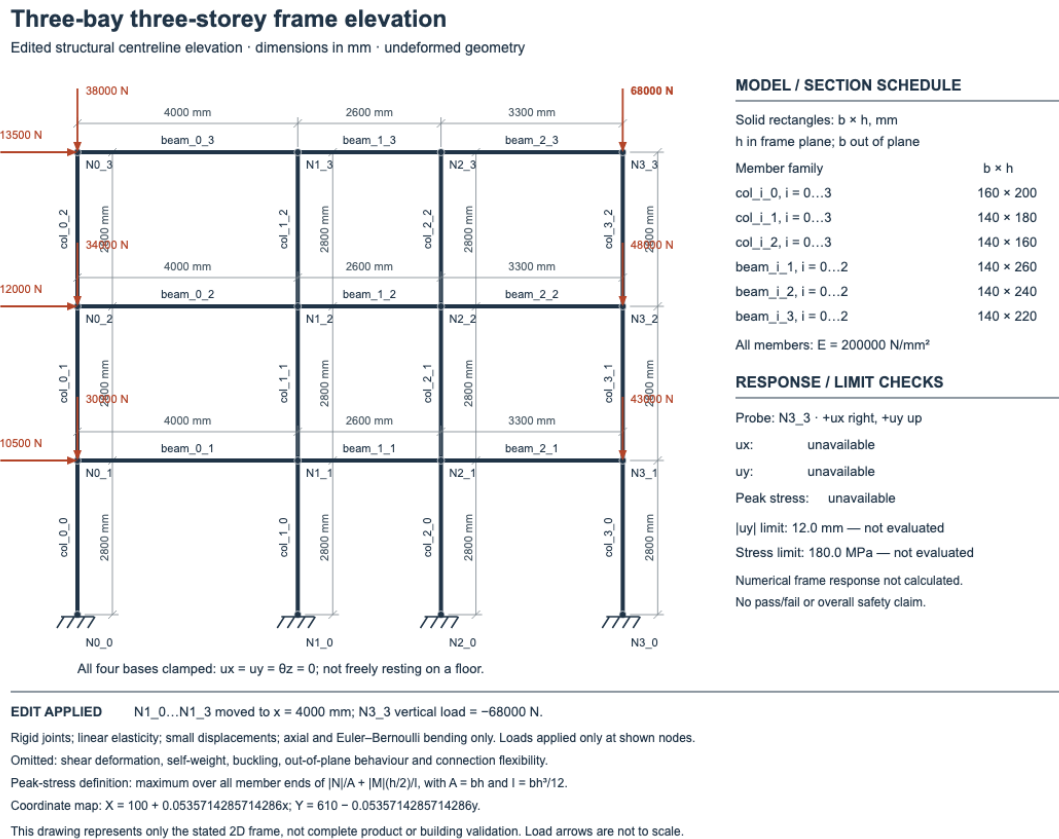
What makes this case instructive is how much the model got right on the way to the wrong number. It named the governing location, col\_EH at end E, which is genuinely the maximum. It stated the correct extreme-fibre definition,  $\sigma_{\max} = \max(|N|/A + |M|(h/2)/I)$ . It solved the displacements essentially exactly. It also labelled the panel *estimated* rather than claiming a solver run. Every diagnostic signal was right; only the arithmetic that reaches the page was wrong.

The consequence is not academic. A drawing released in this state condemns a safe structure and triggers a redesign, and no reader could detect the error from the drawing.

**It does not reproduce.** Three fresh samples at the identical protocol returned 20.4 MPa. Under the project's own admission rule — three completed samples all failing — this is not a systematic capability limit. It is a sporadic error, observed once in 42 completed samples.

## 6 Failure case B: a correct drawing that refuses to conclude

The second case is the one that reproduces. As the frame grows, the model keeps drafting to the same standard but stops short of the analysis rather than guessing at it.



**Figure 4:** threebay\_three\_edit, 21 members,  $i = 27$ , unmodified output. The frame, all 21 member dimensions, the applied edit and the modelling notes are complete and correct. The response panel is empty.

### RESPONSE / LIMIT CHECKS

Probe: N3\_3 · +ux right, +uy up

ux: unavailable

uy: unavailable

Peak stress: unavailable

|uy| limit: 12.0 mm — not evaluated

Stress limit: 180.0 MPa — not evaluated

Numerical frame response not calculated.

No pass/fail or overall safety claim.

**Figure 5:** Detail. The model returns **status:** "unavailable", declines both limit checks and states "no pass/fail or overall safety claim". It gave a specific reason: the edited frame stiffness solution and member-end force recovery were not performed.

This is the behaviour that *does* reproduce, and it is tied to frame size.

| Members | Samples | Correct | Wrong | Declined |
|---------|---------|---------|-------|----------|
| 3–18    | 35      | 34      | 1     | 1 (3 %)  |
| 20–21   | 7       | 3       | 0     | 4 (57 %) |

The contrast is significant at  $p = 0.0015$  (two-sided Fisher exact); the 57 % rate carries a 95 % Wilson interval of 25–84 %. Token budget is not the cause: declines finish cleanly after 6,600–8,100 completion tokens against a 32,000 cap, while successful answers on the same instances use more. It is a boundary in willingness to attempt an unaided indeterminate solve, not in drafting — the drawing is always finished.

## 7 What the two cases show together

Accuracy does not decay with complexity; willingness does. In all 36 samples where a number was given it was within tolerance, including at 21 members and  $i = 27$ , where the model returned 17.7 MPa against a reference of 17.6837. The model is either right or silent, with one observed exception.

That inverts the premise the programme was built on. The expectation was that the model would draw plausibly and be numerically wrong, making verification a lie detector. Measured, the model is numerically right whenever it commits and says so when it cannot. The verification layer’s primary value is therefore **completion** — supplying the solve the model declines to attempt, so the drawing is finished rather than rejected — with silent-error detection as a secondary, still necessary, guard that Figure 3 shows is not hypothetical.

**What this says about the architecture.** The measurement lands exactly on the seam in Figure 1. Geometry was never the problem: across all 42 samples the frames, dimensions and joints were correct at every size, in both generation and editing. The failures were confined to the numbers — wrong once, refused five times. That is the empirical case for letting a model plan and edit while a solver and a deterministic exporter own the quantities, rather than asking the model to compute and draw them together. It also sets the priority: the exporter and verifier stages are not defensive scaffolding around a weak drafter, they are the stages that carry the only part of the job that actually breaks.

One further caution, from our own harness: three results were initially recorded as model failures for omitting `mm` on individual dimension labels, when each drawing declared its units once in the notes. Stating units once is standard drafting practice; the checker was penalising correct work. Two of the four leads chased in this study were artefacts of measurement rather than of the model.

## 8 Scope and reproduction

Planar rigid-jointed frames, linear elastic, small displacement, nodal loads, rectangular sections; self-weight, shear deformation, buckling and connection flexibility are outside the model. Scoring audits tagged geometry and numbers, not rendering quality, layers or title-block convention. All runs used `gpt-6-astra` at medium reasoning effort.

Two limits on the data itself. The boundary is bracketed rather than resolved: 18 members passed twice, 20 and 21 declined in 4 of 7 samples, and member count cannot be separated from degree of indeterminacy because they rise together in this generator. And one instance, `twobay_four_generate` at 20 members, rests on its single screened sample — its confirmation runs were cut off when API credits were exhausted, and are recorded as excluded rather than counted as failures.

**Reproducing.** From cvpr2027/ with PYTHONPATH=src:

```
../.venv/bin/python scripts/eng_frame_bench.py build
../.venv/bin/python scripts/eng_frame_bench.py verify
../.venv/bin/python scripts/cad_astra_benchmark.py run --data data/eng-frame-bench \
    --output runs/<fresh> --samples 3
../.venv/bin/python scripts/eng_frame_report.py
```

Full study, statistics and limits: reports/ENG\_FRAME\_FAILURE\_BOUNDARY\_20260922.md and docs/latex/eng\_frame\_failure\_study.tex.